

FinanceAI

Intelligent Personal Finance & Risk Analysis System

FULL SEMINAR PRESENTATION SCRIPT

Hinglish Version | 4-Member Team | 12-15 Minute Seminar Flow

Member	Role
Ajit Kumar	Team Leader - Full-Stack & Integration Lead
Ajeet Kumar	Personal Finance & Data Analytics Lead
Sonu Kumar	Credit Stress ML & Explainability Lead
Pranav Kr Mishra	Loan Risk ML, Testing & Documentation Lead

Recommended flow: Ajit -> Ajeet -> Sonu -> Pranav -> Ajit

Use this PDF as a rehearsal script. Speak naturally - do not read every word mechanically.

How to Use This Script in Seminar

- Total target duration: around 12-15 minutes including a short live demo.
- Do not memorize word-by-word. Remember the idea of each paragraph and speak naturally.
- When changing speaker, use the transition line given at the end of the section.
- During live demo, avoid typing long data manually. Keep sample inputs prefilled.
- Always mention that Credit Stress and Loan Risk are experimental analysis modules, not approval/rejection engines.

Part 1 - Ajit Kumar: Opening, Problem, Solution and Architecture

Slide 1 - Title: FinanceAI

Speaker: Ajit Kumar

Good morning respected sir, ma'am and everyone present here.

My name is Ajit Kumar, and I am the team leader of our project.

Today, our team is going to present our final project, FinanceAI - Intelligent Personal Finance and Risk Analysis System.

Our team consists of four members: Ajit Kumar, Ajeet Kumar, Sonu Kumar, and Pranav Kr Mishra.

FinanceAI is basically a Flask-based integrated financial analytics application. Is project me humne Personal Finance analysis, Machine Learning, authentication, database, APIs, analysis history aur explainable risk-analysis modules ko ek single application ke andar integrate kiya hai.

Hamare project ka main purpose sirf prediction karna nahi hai. Hamara goal financial data ko understandable, explainable aur useful insights me convert karna hai.

Slide 2 - Meet the Team

Speaker: Ajit Kumar

Before moving further, I would like to introduce our team and responsibilities.

Main Ajit Kumar, Team Leader hoon, aur maine mainly overall architecture, Flask backend, authentication, database integration, APIs aur final module integration par work kiya hai.

Ajeet Kumar Personal Finance and Data Analytics part explain karenge.

Sonu Kumar Credit Stress Machine Learning and Explainability module handle kar rahe hain.

And Pranav Kr Mishra Loan Risk, testing and documentation module explain karenge.

Testing, debugging, documentation review aur final presentation ham sabki shared responsibility hai.

Slide 3 - Problem Statement

Speaker: Ajit Kumar

Now I would like to explain the problem that motivated us to build FinanceAI.

Aaj almost har person ke paas financial information hoti hai, jaise monthly income, expenses, savings, loan payment, investment aur credit-related data. But problem ye hai ki raw financial numbers ko samajhna easy nahi hota.

For example, agar kisi ki salary fifty thousand rupees hai, to sirf salary dekhkar hum ye decide nahi kar sakte ki uski financial condition healthy hai ya nahi. Hume expenses, debt, emergency fund, savings aur investment bhi dekhna padega.

Second problem ye hai ki financial risk models kabhi-kabhi black-box hote hain. Model ek score de deta hai, but user ko clearly nahi pata hota ki result ka meaning kya hai, kis basis par prediction hua, aur model ki limitations kya hain.

So our problem statement was: Can we build one system which combines explainable financial-health analysis, machine-learning risk analysis and a complete application backend?

Isi problem ko solve karne ke liye humne FinanceAI develop kiya.

Slide 4 - FinanceAI Solution Overview

Speaker: Ajit Kumar

FinanceAI ko humne three main analytical modules me divide kiya hai.

First module is Personal Finance. Ye user ko zero to hundred ka Financial Health Score, component breakdown aur personalized recommendations deta hai.

Second module is Credit Stress. Isme humne Logistic Regression based machine-learning benchmark use kiya hai.

Third module is Loan Risk. Isme Extra Trees based experimental risk-ranking model use kiya gaya hai.

Aur in tino modules ko humne ek Flask application ke through integrate kiya hai. Application me authentication, database, history, APIs aur offline FinanceAI Assistant bhi available hai.

Slide 5 - System Architecture

Speaker: Ajit Kumar

Now let us understand how FinanceAI works technically.

Sabse pehle user browser ya API ke through application ko access karta hai. User request Flask application par aati hai. Flask layer authentication aur input validation handle karti hai.

Uske baad request service layer me jati hai. Service layer decide karti hai ki Personal Finance engine, Credit Stress model ya Loan Risk model me se kaunsa module call karna hai.

Personal Finance module transparent rule-based scoring engine use karta hai. Credit Stress Logistic Regression pipeline use karta hai. Loan Risk Extra Trees pipeline use karta hai.

Final result SQLite ya MySQL database me save kiya ja sakta hai. User later History page ke through past results dekh sakta hai. Assistant latest saved Personal Finance result ko explain bhi kar sakta hai.

So overall architecture hai: User -> Flask -> Service Layer -> Analysis Engine or ML Model -> Database -> Result and History.

Transition: Ab Personal Finance module ko detail me explain karne ke liye main Ajeet ko invite karta hoon.

Part 2 - Ajeet Kumar: Personal Finance Module

Slide 6 - Personal Finance Module

Speaker: Ajeet Kumar

Thank you Ajit.

Now I am going to explain our first and primary user-facing module, which is Personal Finance.

Is module ka purpose hai user ki current financial condition ko understandable form me represent karna.

User kuch important inputs provide karta hai, such as monthly income, monthly expenses, debt-to-income ratio, loan payment, investment amount, emergency fund, credit score, budget goal, and actual savings.

In values ke basis par FinanceAI ek Financial Health Score from zero to hundred calculate karta hai.

Important point ye hai ki ye score black-box Machine Learning prediction nahi hai. Ye transparent rule-based system hai. Isliye hum clearly explain kar sakte hain ki total score kaunsa component kitna contribute kar raha hai.

Personal Finance module ka output sirf score nahi hai. User ko health category, ratios, component breakdown aur personalized recommendations bhi milti hain.

Slide 7 - How the 100-Point Health Score is Built

Speaker: Ajeet Kumar

Now let us understand how the hundred-point score is calculated.

Health Score six components se milkar banta hai.

First is Cash Flow - maximum 25 points.

Second is Debt Health - maximum 20 points.

Third is Emergency Fund - maximum 20 points.

Fourth is Savings Progress - maximum 15 points.

Fifth is Investment - maximum 10 points.

And sixth is Credit - maximum 10 points.

So total becomes 25 plus 20 plus 20 plus 15 plus 10 plus 10, which is equal to 100 points.

After calculating the score, system user ko health bands me classify karta hai. 85 to 100 Excellent, 70 to 84 Good, 55 to 69 Moderate, 40 to 54 Weak, and below 40 Critical.

Example integration test me score 77 out of 100 aaya, jo Good category hai.

Along with score, recommendation engine weak areas ko identify karta hai. Agar expenses high hain to spending reduce karne ki recommendation, emergency fund low hai to emergency savings build karne ka suggestion, aur debt-to-income ratio high hai to additional debt avoid karne ka recommendation mil sakta hai.

Live Demo - Personal Finance

Speaker: Ajeet Kumar

Now I would like to show a small live demo.

Sabse pehle hum Personal Finance page open karte hain. Yahan user monthly income, expenses, debt-to-income ratio, loan payment, investment, emergency fund, credit score, budget goal aur savings enter karta hai.

Now I click on Analyze.

System Financial Health Score generate karta hai. Yahan health level show ho raha hai. Neeche component-wise score show ho raha hai. Aur recommendations section me prioritized actions show ho rahe hain.

After saving the result, ye analysis History page me available ho jata hai.

So Personal Finance module is explainable, transparent and user-focused.

Transition: Ab Credit Stress Machine Learning module Sonu explain karenge.

Part 3 - Sonu Kumar: Credit Stress Machine Learning

Slide 8 - Credit Stress Model

Speaker: Sonu Kumar

Thank you.

Now I am going to explain our Credit Stress Machine Learning module.

In this module, we have compared different classification models. After model comparison, the Logistic Regression model was selected as the champion.

But before discussing metrics, one important point is clear: it is necessary. The Credit Stress module's target is `financial_stress_flag`.

The actual credit default is not the target. It is a rule-based financial-stress proxy target.

To avoid target leakage, we intentionally used only five strict predictors: Age, Dependents, Occupation, City Tier, and Desired Savings Percentage.

Income and expense features directly construct the target, so we excluded them from the strict ML predictor set to make the decision model scientifically safer.

Slide 8 - Credit Stress Performance

Speaker: Sonu Kumar

Now coming to the performance.

The champion model was Logistic Regression.

Test ROC-AUC is approximately 0.854. PR-AUC is approximately 0.233.

The default threshold is initially 0.50. In STEP 7, we performed threshold tuning, and the recommended threshold became 0.68.

At this tuned threshold, Recall is approximately 82.2 percent, Precision is approximately 25.1 percent, and F1 score is approximately 0.385.

Top-decile lift is approximately 3.82 times.

The dataset is imbalanced, with positive stress cases around 6.9 percent. Therefore, we place more importance on PR-AUC, Recall, Precision, and F1 score.

Slide 9 - Explainability and Threshold

Speaker: Sonu Kumar

Logistic Regression choose karne ka ek additional advantage hai explainability.

Logistic Regression coefficients provide karta hai. Agar coefficient positive hai, to wo feature model ka stress score increase karta hai. Agar coefficient negative hai, to model score decrease hota hai.

Hum odds ratio bhi calculate kar sakte hain using: $\text{Odds Ratio} = \exp(\text{coefficient})$.

Isse prediction ka direction understand karna easier hota hai.

STEP 7 me threshold tuning ka purpose false positives aur missed positives ke beech better balance achieve karna tha. Threshold 0.68 par precision improve hui aur recall abhi bhi 80 percent se above raha.

But ek limitation clearly mention karna important hai: This model does not predict actual credit default. It predicts a financial-stress proxy. So FinanceAI is model ko credit-card approval ya rejection ke liye use nahi karta.

Optional Live Demo - Credit Stress

Speaker: Sonu Kumar

Ab hum Credit Stress page open karte hain.

Yahan sirf strict predictors input karte hain: Age, Dependents, Occupation, City Tier and Desired Savings Percentage.

Submit karne ke baad system experimental stress score aur classification show karta hai.

Saath me warning bhi clearly show hoti hai ki ye real default probability nahi hai.

Transition: Ab Loan Risk module Pranav explain karenge.

Part 4 - Pranav Kr Mishra: Loan Risk Module

Slide 10 - Loan Risk Model

Speaker: Pranav Kr Mishra

Thank you.

Now I am going to explain the Loan Risk module.

Loan Risk dataset me positive risky cases bahut kam the. Ye severe class imbalance create karta hai.

Isliye agar hum sirf accuracy dekhte, to misleading result mil sakta tha. For example, model almost sabko safe predict karke bhi high accuracy show kar sakta hai.

Isliye humne PR-AUC, ROC-AUC, Recall, F1 aur MCC jaise metrics use kiye.

Different models compare karne ke baad Extra Trees champion model select hua.

Test ROC-AUC approximately 0.6487 tha. Test PR-AUC approximately 0.0326 tha. STEP 7 me recommended threshold approximately 0.232 select hua.

Loan Risk Limitation and Safety

Speaker: Pranav Kr Mishra

However, model ki limitation important hai.

Positive cases rare hain. Historical observation window limited hai. Aur tuned threshold ke baad bhi model performance real lending use ke liye strong enough nahi hai.

Isliye humne system me Loan Risk ko clearly experimental risk-ranking model ke roop me label kiya hai.

FinanceAI Loan Risk model ko automatic approval or rejection ke liye use nahi karta.

Ek aur important safety feature ye hai ki Loan Risk API complete application-time feature vector demand karti hai.

Agar required input missing hai, system silently median ya default value use karke misleading prediction nahi karta. Instead, incomplete input ko reject karta hai.

Optional Live Demo - Loan Risk

Speaker: Pranav Kr Mishra

Now I will show a small demo.

Loan Risk page par sample input load karte hain. System borrower-related application-time values ko model pipeline me pass karta hai.

Output me experimental risk score show hota hai. Threshold ke comparison me class label generate hota hai.

Saath me warning bhi show hoti hai ki ye autonomous lending decision nahi hai.

So the important point is: FinanceAI risk analysis demonstrate karta hai, but decision automate nahi karta.

Transition: Ab final backend features, future scope aur conclusion ke liye main presentation wapas Ajit ko hand over karta hoon.

Part 5 - Ajit Kumar: Backend, User Flow, Responsible Use and Conclusion

Slide 11 - Final Application Features

Speaker: Ajit Kumar

Thank you Pranav.

Ab main explain karna chahunga ki FinanceAI sirf Machine Learning notebooks ka collection nahi hai.

Humne complete working Flask application develop kiya hai.

Application me user registration, login and logout, password change, secure password hashing, CSRF protection, basic login rate limiting, analysis history, CSV export aur offline FinanceAI Assistant available hain.

Har authenticated user ki analysis history separately maintain hoti hai.

Project JSON APIs bhi expose karta hai.

So our final project combines: Full Stack Development + Data Analytics + Machine Learning + Database + Security + Explainability.

Slide 12 - Database, Security and APIs

Speaker: Ajit Kumar

For database, FinanceAI supports two options.

First is SQLite. SQLite zero-setup hai, isliye instant local demo ke liye useful hai.

Second is MySQL, which can be used with XAMPP.

Database me user records aur analysis history persist hoti hai.

Security side par humne password hashing, CSRF protection, session authentication and basic login throttling implement kiya hai.

Project APIs me examples hain:

GET /api/v1/status

POST /api/v1/personal-finance

POST /api/v1/credit-stress

POST /api/v1/loan-risk

GET /api/v1/history

Ye APIs project ko future integration ke liye useful banati hain.

Slide 13 - End-to-End User Journey

Speaker: Ajit Kumar

Complete user flow ko agar simple terms me samjhein, to six main steps hain.

First, user registers or logs in.

Second, user dashboard open karta hai.

Third, required analysis module select karta hai.

Fourth, input provide karke analysis run karta hai.

Fifth, system result show karta hai aur relevant result history me save hota hai.

And sixth, user Assistant ke through apne Personal Finance result ko simple language me samajh sakta hai.

So humne complete user journey ko simple, integrated aur reusable rakha hai.

Combined Live Demo Flow

Speaker: Ajit Kumar

Ab hum complete application ka short live flow show karte hain.

First, this is our FinanceAI home page. Yahan project overview aur team information available hai.

Now I will open Login page and demo user login karte hain.

After login, Dashboard open hota hai. Dashboard se three analytical modules accessible hain: Personal Finance, Credit Stress and Loan Risk.

Ab hum History open karte hain, jahan previous analysis records available hain.

Next, FinanceAI Assistant open karte hain. Agar user asks, 'What does my score mean?', Assistant latest saved Personal Finance result ko educational way me explain karta hai.

Finally, I will open /api/v1/status. Yahan backend system status JSON format me available hai.

This demonstrates that FinanceAI has both browser UI and API support.

Slide 14 - Responsible Use

Speaker: Pranav Kr Mishra + Ajit Kumar

FinanceAI clearly distinguishes what the system can do and what it cannot do.

FinanceAI can: explain personal financial health, generate educational recommendations, demonstrate experimental risk ranking, and show model metrics and limitations.

But FinanceAI does not: approve or reject a credit card, approve or reject a loan, claim that proxy labels are real defaults, or provide professional financial advice.

Humne intentionally model limitations ko hidden nahi rakha.

Hamare liye responsible analytics ka meaning hai: Useful output plus honest boundaries.

Slide 15 - Future Scope

Speaker: Ajit Kumar

Now coming to future scope.

Current project academic and portfolio stage par complete hai.

Future me FinanceAI ko real-world data ke saath improve kiya ja sakta hai.

Real bank or UPI transaction integration add ki ja sakti hai. Credit Stress aur Loan Risk ke liye real verified repayment and default outcomes collect kiye ja sakte hain. Longer historical time-series data use karke better temporal validation perform ki ja sakti hai.

Fairness and bias testing add ki ja sakti hai. Risk probabilities ko properly calibrate kiya ja sakta hai. Application ko secure cloud environment par deploy kiya ja sakta hai. Mobile application ya Progressive Web App develop ki ja sakti hai.

Future FinanceAI Assistant ko bhi advanced user-approved financial context ke saath improve kiya ja sakta hai.

But our first priority in future would be: Better real-world data before building more complex models.

Slide 16 - Final Conclusion

Speaker: Ajit Kumar

To conclude our presentation,

FinanceAI is not only a Machine Learning project. It is an integrated financial analytics system.

Through this project, we worked on data understanding, data cleaning, EDA, feature engineering, Machine Learning, model evaluation, threshold tuning, explainability, Flask backend, authentication, database persistence, APIs, security, testing, and responsible communication of model limitations.

Our Personal Finance module provides an explainable Financial Health Score and recommendations.

Our Credit Stress module demonstrates Logistic Regression based financial-stress classification.

Our Loan Risk module demonstrates experimental Extra Trees risk ranking.

And all modules are integrated into one working Flask application.

Most importantly, we do not hide the limitations of our models.

FinanceAI demonstrates that a good AI or ML system is not only about getting a prediction. It is also about making the result understandable, testable, secure and responsibly communicated.

Through this project, we learned both Machine Learning and real application integration.

On behalf of our entire team - Ajeet Kumar, Sonu Kumar, Pranav Kr Mishra and myself Ajit Kumar - thank you for giving us the opportunity to present our project.

Thank you. We are ready for your questions.

Important Viva / Seminar Questions and Ready Answers

Project ek line me batao.

FinanceAI ek Flask-based integrated financial analytics platform hai jo Personal Finance Health Score, explainable recommendations, experimental Credit Stress aur Loan Risk analysis, authentication, database, APIs aur history ko ek system me combine karta hai.

AI / Machine Learning kahan use hua hai?

Machine Learning Credit Stress aur Loan Risk modules me use hua hai. Credit Stress me Logistic Regression aur Loan Risk me Extra Trees use hua hai. Personal Finance module intentionally transparent rule-based hai, because health score ko explainable rakhna hamara design goal tha.

Credit Stress me accuracy kitni hai?

Positive class imbalanced hai, isliye accuracy primary metric nahi hai. Test ROC-AUC approximately 0.854 aur PR-AUC approximately 0.233 tha. Tuned threshold 0.68 par Recall around 82.2 percent aur F1 around 0.385 tha.

Loan model weak kyu hai?

Loan Risk dataset me positive risky cases bahut rare hain aur historical maturity limited hai. Is wajah se rare-event performance weak hai. Isliye model ko production-ready claim nahi kiya aur sirf experimental risk-ranking module ke roop me use kiya.

Why Logistic Regression?

Logistic Regression ne Credit Stress module me best cross-validation PR-AUC diya aur iska major advantage explainability hai. Coefficients aur odds ratios ke through prediction direction ko samjha sakte hain.

Why Extra Trees?

Multiple models compare kiye gaye aur training cross-validation ke basis par Extra Trees Loan Risk ka best candidate tha. Final holdout performance limited thi, isliye ise experimental hi rakha.

Database kya use kiya hai?

FinanceAI SQLite ko default instant-demo database ke roop me support karta hai, aur MySQL/XAMPP optional backend bhi available hai. Database user accounts aur per-user analysis history persist karta hai.

Security features kya hain?

Password hashing, session authentication, CSRF protection, basic login rate limiting aur user-specific history access implement kiya hai. Production banking system ke liye additional security controls required honge.

Why PR-AUC?

Positive classes imbalanced hain. PR-AUC minority positive class ke ranking performance ko accuracy se better reflect karta hai, especially jab positive class rare ho.

Credit Stress actual default model kyu nahi hai?

Kyuki target financial_stress_flag ek rule-based proxy hai, observed repayment default, delinquency ya charge-off nahi.

FinanceAI real loan approve karega?

Nahi. Credit Stress aur Loan Risk dono experimental analytical modules hain. FinanceAI automatic approval/rejection decisions nahi leta.

Flask kyu use kiya?

Flask lightweight hai, Python ML pipelines ke saath easily integrate hota hai, sessions aur APIs support karta hai, aur portfolio-level full-stack application ke liye suitable hai.

Last-Minute Seminar Checklist

- Laptop charger, mouse and presentation clicker (if available) ready rakho.
- PPT, project folder and this PDF offline copy me rakho.
- python app.py presentation se pehle run karke <http://127.0.0.1:5000> test kar lo.
- Demo user pehle se create rakho.
- Personal Finance aur Loan Risk sample input prefilled rakho.
- Browser me Home, Dashboard, History, Assistant and /api/v1/status tabs ready rakho.
- Internet par depend mat karo - demo local rakho.
- Har member apne transition sentence ko rehearse kare.
- Metrics ko memorize karne ke bajay unka meaning samjho.
- Agar live demo fail ho, calmly PPT screenshots/flow se continue karo.

Presentation order: Ajit -> Ajeet -> Sonu -> Pranav -> Ajit

Final line: Thank you. We are ready for your questions.